

CarbonCheck

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Project Objectives

- Problem Statement
- Project Overview – Introduction
- End Users
- Wow Factor in Project
- Modelling/Block Diagram/Flow of Project
- Result/outcomes
- Conclusion
- Future Perspective



Problem Statement

Rising Carbon Emissions

Carbon emissions from daily activities such as transportation, electricity usage and consumption habits continue to contribute significantly to climate change.

Limited User Awareness

Most individuals are unaware of how their day-to-day decisions affect their personal carbon footprint.

Need for an Intelligent Solution

There is a need for an AI-based system that can estimate carbon emissions more accurately and provide personalized suggestions to encourage sustainable living.

Lack of Personalized Guidance

Existing carbon footprint calculators generally provide generic results and do not offer recommendations based on individual lifestyle patterns.

Static Calculation Methods

Many existing solutions rely on fixed formulas and are unable to adapt to variations in user behaviour and consumption habits.

Project Overview - Introduction

- CarbonCheck is an AI-powered platform designed to help users understand and reduce their carbon footprint through data-driven analysis.
- The system collects information related to transportation habits, electricity usage and lifestyle choices to estimate an individual's carbon emissions.
- Using machine learning techniques, the platform classifies users into different emission categories and generates personalized recommendations for adopting more sustainable practices.
- The project aims to increase environmental awareness and encourage small lifestyle changes that can contribute to long-term sustainability goals.



End Users

Environmentally Conscious Individuals

Individuals who want to better understand the environmental impact of their daily activities and identify practical ways to reduce their carbon footprint.

Families and Households

Families can use the platform to analyse electricity consumption, transportation habits and lifestyle choices in order to adopt more sustainable living practices.

Students and Educational Institutions

Educational institutions can use the platform for awareness programs, sustainability projects and environmental education initiatives.

Organizations and Companies

Companies can encourage employees to monitor their carbon footprint and promote environmentally responsible workplace practices.

Communities and Sustainability Programs

Government initiatives, NGOs and local communities can use the platform to support campaigns related to climate action and sustainable development.

Researchers and Policy Makers

The generated insights and consumption patterns can help researchers and policy makers better understand behavioural trends related to carbon emissions.

Wow Factors in Project

AI-Based Prediction

Uses machine learning to predict emission levels instead of relying only on fixed calculations.

Personalized Suggestions

Provides recommendations based on the user's lifestyle and consumption habits.

Interactive Dashboard

Displays carbon footprint data using charts and visual analytics.

SDG Aligned Solution

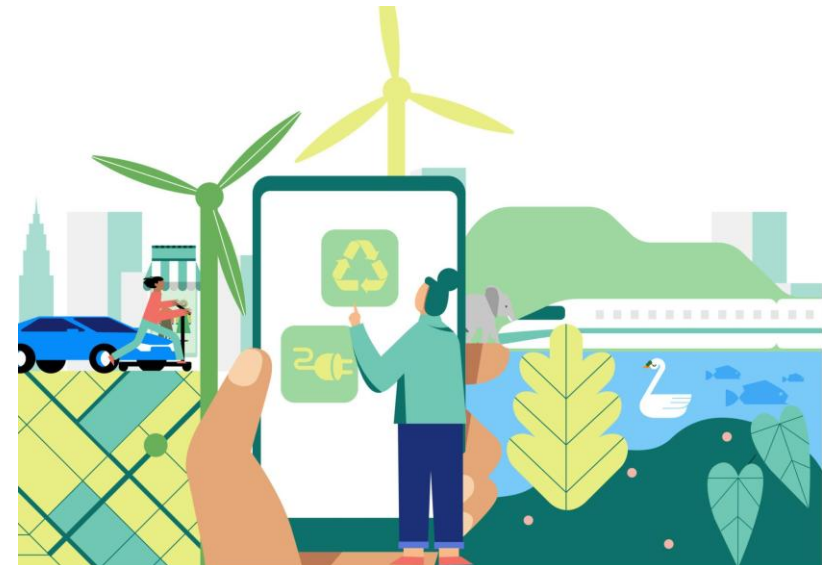
Supports global sustainability goals and promotes environmental awareness.

Instant Results

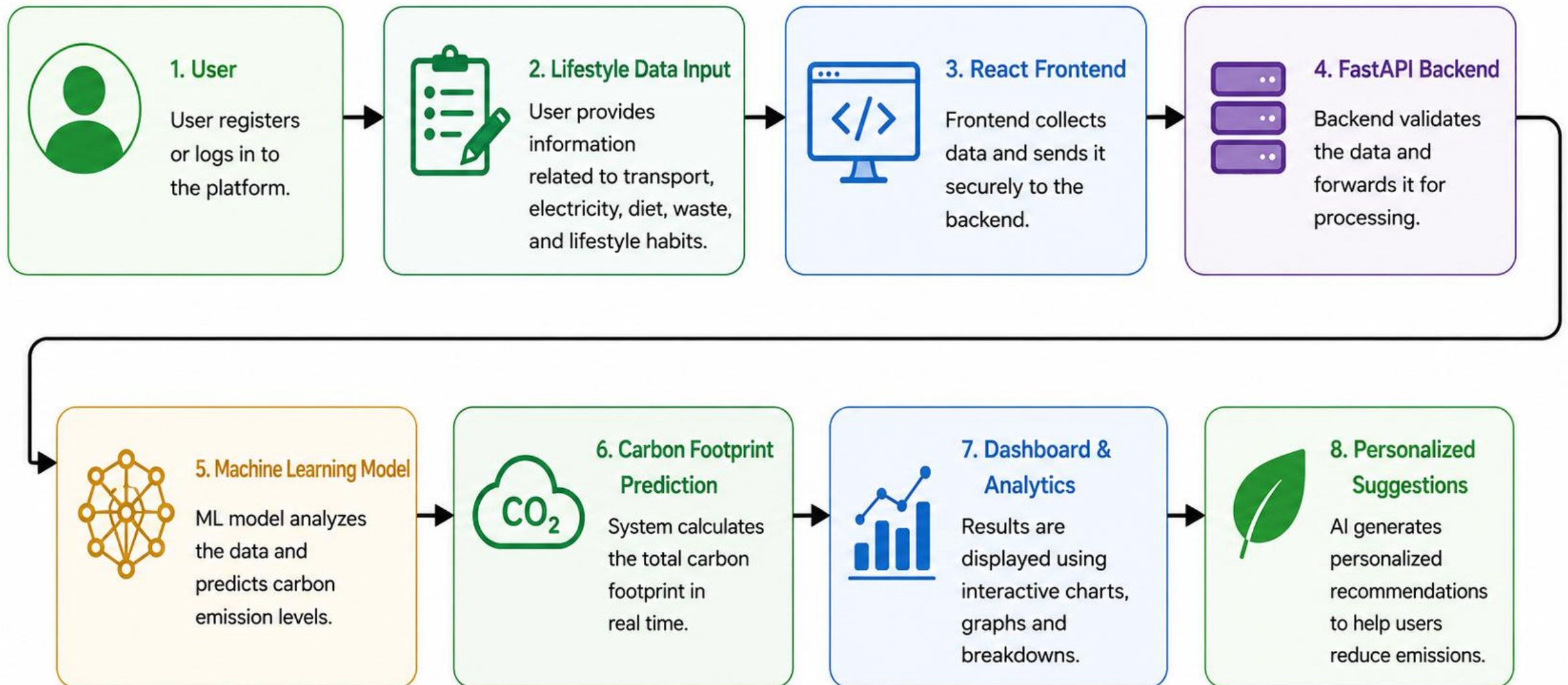
Calculates and displays emission estimates in real time.

User-Friendly Design

Simple and responsive interface accessible across multiple devices.



Modelling/Block Diagram/Flow of Project



Technology Stack



Frontend

- React.js
- Tailwind CSS
- Framer Motion
- React Router
- Recharts



Backend

- Python
- FastAPI



Database

- SQLite



AI / ML

- Scikit-Learn
- Pandas
- NumPy

Result/Outcomes



AI Predicts Carbon Footprint Successfully

The model reliably predicts emission levels from user lifestyle data.



Interactive Dashboard

Users can visualize their emissions using graphs, charts and detailed breakdowns.



AI-Based Emission Classification

The machine learning model is capable of categorizing users into different emission levels such as Low, Medium and High.



Responsive User Interface

The platform performs smoothly across desktop and mobile devices.



Fast Processing

FastAPI backend returns predictions with minimal latency.



Personalized Recommendations

The platform generates suggestions tailored to the user's habits to help reduce carbon emissions.

Conclusion

- CarbonCheck successfully combines artificial intelligence and sustainability analytics to estimate individual carbon footprints and promote environmentally responsible behaviour.
- Unlike traditional carbon calculators, the platform provides personalized recommendations and interactive visualizations based on user lifestyle patterns and consumption habits.
- The project demonstrates how machine learning can be used to increase environmental awareness and encourage sustainable decision-making in everyday life.



Future Perspective

Mobile App

Develop a dedicated mobile application for easy carbon tracking anytime and anywhere.

IoT Integration

Connect smart meters and IoT devices to automatically collect energy consumption data.

Google Maps API

Use location data to estimate transportation emissions more accurately.

Cloud Deployment

Deploy the platform on cloud infrastructure for better scalability and accessibility.

Carbon Credit System

Reward users with carbon credits for adopting sustainable habits and reducing emissions.

Smart Home Integration

Integrate with smart home devices to monitor electricity usage automatically.

Renewable Energy Monitoring

Track solar energy production and renewable energy consumption patterns.

Deep Learning

Use advanced deep learning models to improve prediction accuracy and recommendation quality.

Thank You

Github link-

<https://github.com/adarsh-agent/CarbonEmission>

Video (google drive link)-

<https://drive.google.com/drive/folders/1mtK19u4uD1oUwWlgYRG4HdL16mlegEve?usp=sharing>